INCH - POUND

MS20029F
20 May 2005
SUPERSEDING
MS20029E
20 December 1967

DETAIL SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, JAM NUT MOUNTING, SOLDER TYPE, BAYONET COUPLING, CLASS P, SERIES I

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 7 December 1998.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27599.

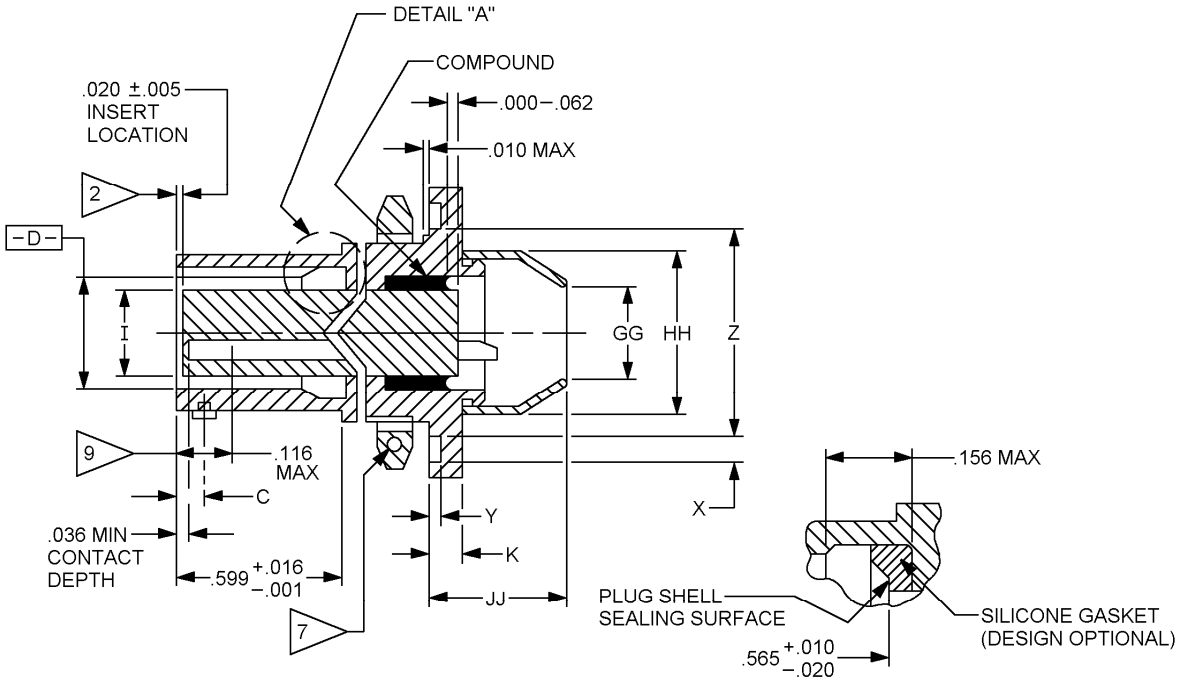
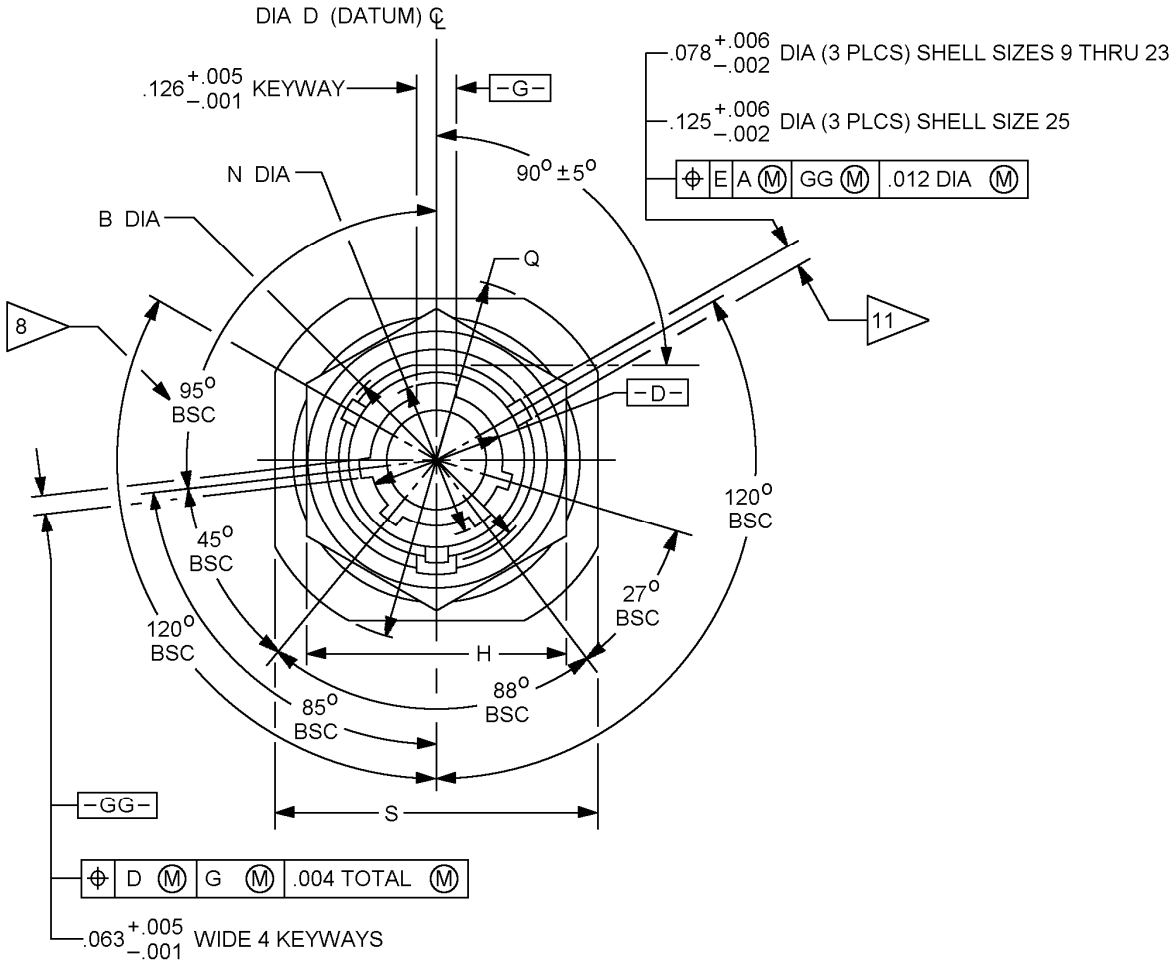


FIGURE 1. Receptacle, jam nut mounting.

AMSC N/A

FSC 5935

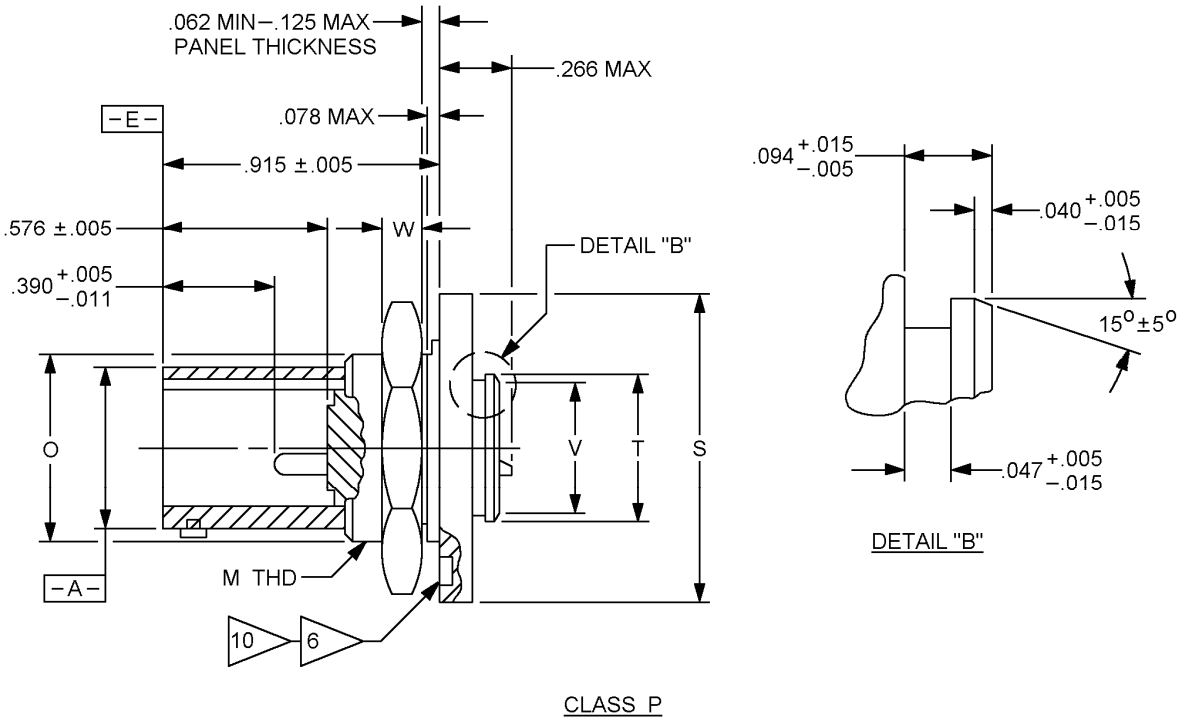
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CLASS P

FIGURE 1. Receptacle, jam nut mounting - Continued.

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CLASS P

NOTES:

1. Dimensions are in inches. Metric equivalents are in parenthesis and given for information only.
2. Insert front surface shall be flat within .005 T.I.R.
3. Diameters A, D, M, Q and Z shall be concentric within .015 T.I.R.
4. Diameters D and I shall be concentric within .005 T.I.R. at MMC.
5. Diameter A with respect to diameter B, and diameter D with respect to diameter N, shall be concentric at MMC.
6. Sides of groove may have 5° max taper.
7. .062 ± .005 dia., 3 lockwire holes, shell sizes 9 through 25.
8. Normal keyway position. For other keyway positions, see MIL-DTL-27599.
9. The point at which a gauge pin having the same basic diameter as the mating contact and a square face, touches socket contact spring.
10. The gauge features for GG shall be CC smaller than their MMC size.

FIGURE 1. Receptacle, jam nut mounting - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.0254	0.016	0.40	0.078	1.98	0.390	9.90
0.002	0.051	0.020	0.51	0.094	2.39	0.565	14.35
0.005	0.127	0.036	0.91	0.116	2.95	0.576	14.63
0.006	0.152	0.040	1.02	0.125	3.17	0.599	15.21
0.010	0.254	0.047	1.19	0.126	3.20	0.915	23.24
0.011	0.28	0.062	1.57	0.156	3.96		
0.015	0.38	0.063	1.60	0.266	6.76		

FIGURE 1. Receptacle, jam nut mounting - Continued.

TABLE 1. Dimensions.

Shell Size	A dia + .001 - .005	B dia + .000 - .016	C BSC	D dia + .005 - .001	H hex + .017 - .016	I dia ± .005	CC
9	.572 (14.53)	.662 (16.81)	.100 (2.54)	.442 (11.22)	.875 (22.22)	.296 (7.52)	.010 (.254)
11	.700 (17.78)	.810 (20.57)		.570 (14.48)	1.000 (25.40)	.413 (10.50)	
13	.850 (21.60)	.960 (24.38)		.687 (17.45)	1.188 (30.17)	.527 (13.38)	
15	.975 (24.76)	1.085 (27.56)		.812 (20.62)	1.312 (33.32)	.652 (16.56)	
17	1.100 (27.94)	1.210 (30.73)		.937 (23.80)	1.438 (36.52)	.777 (19.73)	
19	1.207 (30.65)	1.317 (33.45)		1.042 (26.46)	1.562 (39.67)	.866 (22.00)	
21	1.332 (33.83	1.442 (36.62)		1.167 (29.64)	1.688 (42.87)	.991 (25.17)	
23	1.457 (37.01)	1.567 (39.80)		1.292 (32.81)	1.812 (46.02)	1.116 (28.34)	
25	1.582 (40.18)	1.692 (42.97)	.109 (2.77)	1.417 (35.60)	2.000 (50.80)	1.241 (31.52)	

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TABLE 1. Dimensions - Continued.

Shell Size	K + .011 - .010	M Thread class 2A, plated, UNEF	N dia + .005 - .006	O + .000 - .010	Q dia ± .016	S ± .016
9	.109 (2.77)	.6875 - 24	.492 (12.50)	.655 (16.64)	1.188 (30.17)	1.062 (26.97)
11		.8125 - 20	.620 (15.75)	.755 (19.17)	1.375 (34.92)	1.250 (31.75)
13		1.0000 - 20	.769 (19.53)	.942 (23.92)	1.500 (38.10)	1.375 (34.92)
15		1.1250 - 18	.894 (22.70)	1.066 (27.07)	1.625 (41.27)	1.500 (38.10)
17		1.2500 - 18	1.019 (25.88)	1.191 (30.25)	1.750 (44.45)	1.625 (41.27)
19		1.3750 - 18	1.124 (28.55)	1.316 (33.42)	1.938 (49.22)	1.812 (46.02)
21	.140 (3.55)	1.5000 - 18	1.249 (31.72)	1.441 (36.60)	2.062 (52.37)	1.938 (49.22)
23		1.6250 - 18	1.374 (34.89)	1.566 (39.77)	2.188 (55.57)	2.062 (52.37)
25		1.7500 - 18	1.499 (38.07)	1.691 (42.95)	2.312 (58.72)	2.188 (55.57)

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TABLE 1. Dimensions - Continued.

Shell Size	T dia + .011 - .000	V dia ± .011	W + .017 - .000	X + .005 - .001	Y dia ± .005	Z dia + .006 - .005
9	.516 (13.10)	.476 (12.10)	.125 (3.17)	.094 (2.38)	.045 (1.14)	.777 (19.73)
11	.642 (16.30)	.602 (15.30)				.965 (24.51)
13	.766 (19.45)	.726 (18.44)				1.090 (27.68)
15	.892 (22.65)	.852 (21.64)				1.215 (30.86)
17	1.018 (25.86)	.978 (24.84)				1.340 (34.04)
19	1.142 (29.07)	1.102 (28.00)		.141 (3.58)	.075 (1.90)	1.449 (36.80)
21	1.268 (32.20)	1.228 (31.20)				1.574 (39.98)
23	1.392 (35.35)	1.352 (34.34)				1.699 (43.15)
25	1.518 (38.55)	1.478 (37.54)				1.824 (46.33)

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TABLE 1. Dimensions - Continued.

Shell Size	N dia Min	HH dia Max	JJ max	Recommended packing “O” ring
9	.317 (8.05)	.625 (15.87)	.688 (17.47)	SAE-AS3578-019
11	.434 (11.02)	.750 (19.05)		SAE-AS3578-022
13	.548 (13.92)	.875 (22.22)		SAE-AS3578-024
15	.673 (17.09)	1.000 (25.40)		SAE-AS3578-026
17	.798 (20.27)	1.125 (28.57)		SAE-AS3578-028
19	.899 (22.83)	1.250 (31.75)	.719 (18.26)	SAE-AS3578-128
21	1.024 (26.01)	1.375 (34.92)	.812 (20.62)	SAE-AS3578-130
23	1.149 (29.18)	1.500 (38.10)		SAE-AS3578-132
25	1.274 (32.36)	1.625 (41.27)		SAE-AS3578-134

REQUIREMENTS:

Dimensions and configurations: See figure 1 and table 1.

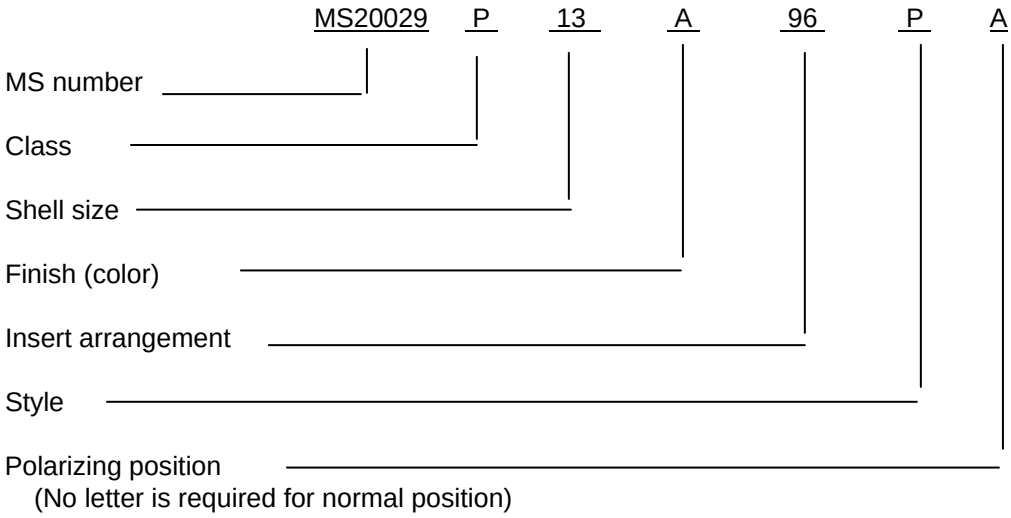
This connector mates with MS20027 and MS20028.

Insert arrangement shall be in accordance with MIL-STD-1560.

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Recommended packing for “O” ring shall be in accordance with SAE-AS3578. Recommended Packing is for dimensions only. Material is optional.

Part or Identifying Number (PIN) example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Reference documents. In addition to MIL-DTL-27599, this document references the following:

MIL-STD-1560
MS20027
MS20028
SAE-AS3578

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Air Force - 99
DLA - IS

(Project 5935-4649-000)

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